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ORIGINAL ARTICLE

Ergonomic differences in mesh placement and mesh fixation between laparoscopic and robotic inguinal hernia repair with mesh

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Abstract

Purpose General differences in surgeon ergonomics between laparoscopic and robotic-assisted inguinal hernia repairs (LIHR vs. RIHR) have been previously studied. However, specific differences in the ergonomics of mesh placement (MP) and mesh fixation (MF) are undetermined. Our aim was to determine if there are differences in the ergonomics of MP and MF between the surgical approaches. We hypothesize that we will identify differences, with the potential for worse ergonomics during LIHR.

Methods Data was collected from fifteen LIHR and fifteen RIHR. All cases were elective, primary inguinal hernias completed by a fellowship-trained minimally invasive surgeon. Surface electromyography (EMG) of four upper extremity muscle groups, including the upper trapezius (UT), anterior deltoid (AD), flexor carpi radialis (FCR) and extensor digitorum (ED), was recorded bilaterally during MP and MF. Muscle activation as a percent of maximum voluntary contraction ($\%MVC_{RMS}$) and muscle fatigue denoted as the median frequency of muscle activations (Fmed) were calculated for each muscle.

Results EMG analysis showed increased $\%MVC_{RMS}$ in LIHR compared to RIHR cases, with significant findings in the left UT, right UT, ED, and FCR for MP and MF and the left FCR during MP. Muscle fatigue was decreased in LIHR compared to RIHR cases, with significant differences in left FCR and right ED and AD.

Conclusion Despite greater muscle activations during LIHR, RIHR had greater muscle fatigue. It is possible that short periods of high muscle activation are ergonomically protective during minimally invasive inguinal hernia repair. Identifying these differences may aid in development of procedure-specific interventions to improve ergonomics.

Keywords EMG analysis · Ergonomics · Inguinal hernia · Mesh placement · Minimally invasive surgery

Introduction

Research has shown that the adoption of minimally invasive surgery (MIS) techniques has led to improved patient outcomes when compared with open techniques, especially with regard to postoperative pain and recovery times [1–3]. In some MIS cases, however, patient benefits may come at the expense of the surgeon, due to increased physical demands from repetitive motions and potentially longer operating times compared to open surgery. Recently, there has been increased attention to, and awareness of, surgeon injuries and fatigue. A 2018 meta-analysis pooling data from over 5,000 surgeons who performed both open and MIS techniques found that 68% of them experienced pain, mostly in their back, neck, and upper extremities. An even higher percentage of them experienced fatigue. Additional reported symptoms in some surgeons included numbness and stiffness. Up to 30% of the time, a surgeon's experience with these symptoms resulted in the surgeon taking their own physical symptoms into account when recommending an operative approach. Importantly, these symptoms were reported to occur more frequently and with higher severity in surgeons performing MIS operations compared to open techniques [4].

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With respect to laparoscopic cases, some studies have found that more than 70% of surgeons report physical complaints during or after the completion of an operation [5, 6]. The development of robotic-assisted surgery led many to believe that there would be decreased physical stress on the surgeon compared to traditional laparoscopic approaches for several reasons, including the ability of the surgeon to be seated at the console rather than standing at the patient's side and elimination of the fulcrum effect of the straight laparoscopic instruments with the joystick hand controls [7]. Nevertheless, reports of surgeon injury and fatigue in robotic-assisted operations have been reported to occur in more than 50% of cases [8]. Further, research on the proposed benefits of robotic-assisted surgery over laparoscopic surgery has been inconclusive, with many conflicting results in the existing literature regarding fatigue and injury patterns comparing the two approaches [7, 9–11].

Inguinal hernia repairs (IHR) are one of the most commonly performed operations worldwide. Over 800,000 are performed annually in the United States alone, incurring over $2.5 billion dollars in health care expenses [12]. The only definitive management for inguinal hernias is surgical repair, and the utilization of mesh drastically improves patient outcomes and decreases recurrence rates [13]. While proper anatomical placement of the mesh is crucial to the success of IHR, it can be a challenging and time-consuming process. The frequency with which IHR are performed makes them an ideal case to study the ergonomics of the operating surgeon. In 2020, the Robotic Inguinal versus Transabdominal Laparoscopic Inguinal Hernia Repair (RIVAL) trial investigated laparoscopic IHR (LIHR) versus robotic-assisted inguinal hernia repairs (RIHR) and found no differences in patient outcomes or in surgeon ergonomics, between the two approaches [14]. One limitation of this study was that it assessed the ergonomics of the procedures as a whole rather than breaking them down into their critical steps. Therefore, it remains unknown if differences in ergonomics specifically related to mesh placement (MP) and mesh fixation (MF) contribute to the overall differences in ergonomics between LIHR and RIHR.

The purpose of this study was to determine if there were differences in surgeon ergonomics during MP and MF between LIHR and RIHR. We hypothesize that differences do exist in the ergonomics of MP and MF between laparoscopic and robotic-assisted approaches with the potential for worse ergonomics in the laparoscopic group.

Materials and methods

Subjects and case selection

A total of four fellowship-trained minimally invasive surgeons were recruited for this study. Each surgeon included in this study had completed > 25 inguinal hernia repairs as the attending surgeon, with mesh either laparoscopically (LIHR group) or through a robotic-assisted (RIHR group) approach prior to inclusion in this study, with the assumption that they had overcome their learning curve [15]. No surgeon with any recent musculoskeletal injuries of the upper limb or back were included. All surgical cases were elective, primary inguinal hernia repairs being performed with the use of mesh. Cases with intra-operative or post-operative complications were excluded from this study. This study was approved by the institution's review board (protocol #572-19-EP), and informed consent was obtained by all participating surgeons at the time of enrollment in the study.

No identifying patient information was collected. However, to ensure similar populations between surgical approaches, patient height, weight, and body mass index (BMI) were collected, as well as age, sex, and laterality of the hernia at the time of the operation.

Surgical approach and operative variables

The surgical approach was determined by the surgeon, and each surgeon was encouraged to perform either LIHR or RIHR with mesh based on their previous training and proficiency level with each respective repair. By allowing the surgeon to choose the operative approach, it was our intent to minimize the ergonomic strain of each task and to mimic their normal operative routine. Robotic-assisted cases were all performed via the transabdominal preperitoneal (TAPP) approach. Laparoscopic cases were all performed via the totally extra-peritoneal (TEP) approach. The mesh used was a heavyweight, macroporous, polypropylene anatomical mesh. Mesh sizes were either 10 × 15 cm or 12 × 16 cm, depending on the patient and hernia size. Mesh fixation was performed at Cooper's ligament, high medially, and high laterally above the anterior superior iliac spine in both approaches. A rapidly absorbable suture was used for MF in the robotic-assisted cases, while a tacking device (either permanent or absorbable) was used for MF in the laparoscopic cases.

Several operative variables were collected, including total operative time, operative time to mesh placement, time required for MP, and time required for MF. The total operative time was measured from the initial incision to the time all wounds were completely closed. MP was measured from the time the surgeon began inserting the mesh into the

patient to the time they were finished placing it in a satisfactory position over the myopectineal orifice. MP occurred just prior to fixation in all laparoscopic cases. For some robotic-assisted cases, the mesh was placed into the abdominal cavity prior to docking of the robot. In these circumstances, the time it took to place the mesh into the abdominal cavity and the time it took to appropriately orient the mesh over the myopectineal orifice were measured separately and combined for MP. MF was recorded as the duration of time the surgeon spent securing the mesh. Operative time to MP was calculated by subtracting the starting time of MP from the starting time of the initial incision.

Electromyographic data collection

Muscle activation and fatigue were measured in the operating room while the surgeon was completing the MP and MF tasks. EMG data were collected utilizing the Trigno wireless surface EMG (Delsys, Inc., MA, USA). Preoperatively, sensors were applied on the skin bilaterally over four upper extremity muscle groups: the upper trapezius (UT), anterior deltoid (AD), flexor carpi radialis (FCR), and extensor digitorum (ED). These muscles were chosen to capture shoulder and forearm muscle activity, and had been previously investigated in other studies [16]. Each sensor has four silver bars that were applied perpendicular to the direction of the muscle fibers. Resting EMG data was collected with the surgeon seated in a relaxed position with their hands in their lap. Following this, maximum voluntary contraction (MVC) was collected for each of the eight muscle groups. This was obtained by having the subject perform isometric contraction of the muscle group against the independent investigator, who did not participate in the operative portion of the case. Application of the EMG sensors, resting EMG data, MVC, and all intraoperative measurements were collected by the same investigator for all thirty operations. Sensor placement did not interfere with surgeon sterility and surgeon sterility did not interfere with EMG data capture, which was confirmed with continuous monitoring that began at the time of resting EMG data collection and concluded at the time that the surgeon broke scrub after skin incision closure.

Raw EMG signals were recorded with a sampling rate of 2000 Hz and processed via MATLAB software (The MathWorks, Inc., Concord, MA, USA) using a band-pass filter of 20–300 Hz to reduce contamination and motion artifact from the readings. The EMG data was then converted as a root-mean-square (RMS) EMG and intraoperative EMG recordings were subtracted from resting EMG data, normalized by MVC, and presented as a percentage of the root mean squares of the MVC ($MVC_{RMS,}$ $\%MVC_{RMS}$) in order to reduce inter-subject variation in muscle activation patterns [16–18]. A higher $\%MVC_{RMS}$ correlates with higher muscle activation in the measured muscle group. Muscle fatigue was assessed using the median frequency (Fmed) of EMG signals during static force exertion, with a lower Fmed correlating with a higher level of muscle fatigue [16, 18].

Surgeon ergonomic measures

The surgeon's body posture was measured using the validated Rapid Upper Limb Assessment (RULA) tool [19]. The RULA was originally developed for use in workplaces where performed tasks are at high risk for upper limb musculoskeletal injury. The RULA has also been utilized in operative settings [14, 20]. Ergonomically correct, according to the RULA, is defined as having an upright spine with the head in a neutral position, upper arms relaxed at the side with elbows at 90°, and no bend or twists at the wrists. Points are added based on the degree of deviation from this score, as well as the amount of repetitive motion and force involved with the task. Two scores, one for the arm and wrist posture and one for the neck, trunk, and leg posture, are calculated, and these are then combined into a single grand score for each side of the body. The same investigator who captured EMG data also completed this assessment for all thirty cases, scoring the surgeon's posture during MP and MF independently. Grand total scores for the RULA tool range from 1 to 7, with higher scores indicating a higher risk for musculoskeletal injury. Scores of 1–2 indicate negligible risk of musculoskeletal injury, scores of 3–4 indicate low risk, and a score of 7 indicates very high risk of injury and should prompt investigation and implementation of changes to improve ergonomics and lower this risk.

Surgeon workload measures

Finally, the NASA Task Load Index (NASA-TLX) was completed immediately following each operation once the surgeon scrubbed out of the case. The NASA-TLX is a six question 20-point Likert style survey and has been previously validated in assessing workload and fatigue in a surgical setting [21]. This assessment measures the mental, physical, and temporal demands of a task as well as effort, frustration, and perceived performance of the surgeon [22, 23].

Outcomes of interest and statistical analysis

Primary outcome measures included differences in muscle activation and fatigue as measured by surface electromyography (EMG) as well as differences in injury risk assessment based on surgeon posture during the tasks as measured by the RULA assessment tool. A secondary outcome measure

was to determine if there are correlations between EMG data and postural assessment scores measured intra-operatively while placing and securing mesh.

Descriptive statistics were used to characterize factors related to patient, surgeon, and procedural demographics in the context of inguinal hernia repair surgeries. Counts and percentages were reported for categorical data, while continuous data were described using means, standard deviations, medians, and interquartile ranges as appropriate. Univariate differences between laparoscopic and robotic-assisted surgical approaches were also evaluated.

A repeated measure multivariate analysis of variance (RMANOVA) was conducted using a generalized linear mixed model (GLMM) to assess differences in surgeon ergonomics. This approach was chosen to control for correlated within-subject measurements in the EMG data. To achieve a residual distribution closer to normality, the percentage of maximal voluntary contraction (%MVC) and the median frequency (FMed) scores were log-transformed prior to analysis. The results were subsequently back-transformed to their original values for interpretation.

Multivariate linear regression analysis using a generalized linear mixed model (GLMM) addressing repeated measurement data was used to evaluate the associations between EMG data and Rapid Upper Limb Assessment (RULA) scores for each ergonomic measurement (muscle activations and fatigue) separately after controlling within-subject measurements between their right side and left side movements. A p-value less than 0.05 was considered statistically significant for all analyses performed using SAS/STAT software, version 9.4 (Cary, NC).

Results

Subjects and cases

Four attending surgeons, two male and two female, were participants in the study. Three surgeons were right-hand dominant. All four surgeons perform operations as if they are right-hand dominant, meaning that instrument set up, dissection, and critical steps were performed the same way amongst all surgeons with the same operative approach. Data was collected from a total of 15 laparoscopic and 15 robotic-assisted inguinal hernia repairs between October 2021 and May 2022. There were no significant differences in the collected patient demographics between the laparoscopic and robotic groups (Table 1). Total procedure time was significantly shorter in the LIHR group compared to the RIHR group (88 min vs. 142.7 min, $p < 0.001$). Operative time to MP was similar in both approaches (LIHR group = 60.6 min, RIHR group = 76.8 min, $p = 0.071$). Time spent on MP was significantly shorter in the RIHR group compared to the LIHR group cases (105.7 s vs. 172.7 s, $p = 0.026$). Conversely, time spent on MF was significantly shorter in the LIHR group compared to the RIHR group cases (59.4 s vs. 357.0 s, $p < 0.001$). Overall time spent manipulating the mesh (MP + MF combined) was significantly shorter in the LIHR group ($p < 0.001$; Fig. 1).

Electromyographic data

After ensuring normalization of the data, each data set was inspected for outliers. A single outlier was identified for muscle fatigue in the right UT for the LIHR group and was excluded. Therefore, for all eight muscles studied, there were 30 measures for muscle activation. For muscle fatigue, there were 30 measures for each of the muscles except for the right UT, which included 29 measures.

Overall, average muscle activations ($\%MVC_{RMS}$) in the LIHR group were higher than in the RIHR group (14.0 vs. 8.7, $p < 0.0001$). For body region, the average $\%MVC_{RMS}$ for ED had the greatest muscle activation (Right = 27.3; Left = 18.3) while FCR was the least activated (Right = 5.8; Left = 5.1). Overall, the average FMed was significantly higher in the LIHR group compared to the RIHR group (74.8 vs. 69.1; $p < 0.0001$), indicating lower muscle fatigue in the LIHR group. For body region, muscle fatigue was lowest in the bilateral ED (right = 102.6; left = 95.1) and highest in the bilateral UT (right = 55.2; left = 53.2; $p < 0.0001$).

During MP, the average $\%MVC_{RMS}$ was significantly higher for the LIHR group compared to the RIHR group in the right ED (39.4 vs. 17.7, $p < 0.0001$), right UT (27.0 vs. 9.7, $p = 0.0004$), right FCR (7.3 vs. 4.2, $p < 0.0001$), left UT (24.8 vs. 9.7, $p = 0.0003$) and left FCR (4.9 vs. 4.5, $p < 0.0001$). There were no significant differences in muscle

Table 1 Patient demographics for laparoscopic and robotic inguinal hernia repairs

Demographic	Laparoscopic ($n = 15$)	Robotic ($n = 15$)	p Value
Gender (n)			
Male	13	14	1.00
Female	2	1	
	Mean, (SD)		
Patient Age (years)	54.6 (17.6)	58.2 (14.9)	0.55
Patient Weight (kg)	87.2 (16.2)	84.7 (18.7)	0.71
Patient Height (cm)	176.9 (8.9)	175.9 (8.2)	0.77
Patient BMI (kg/m^2)	27.7 (3.5)	27.2 (5.2)	0.77

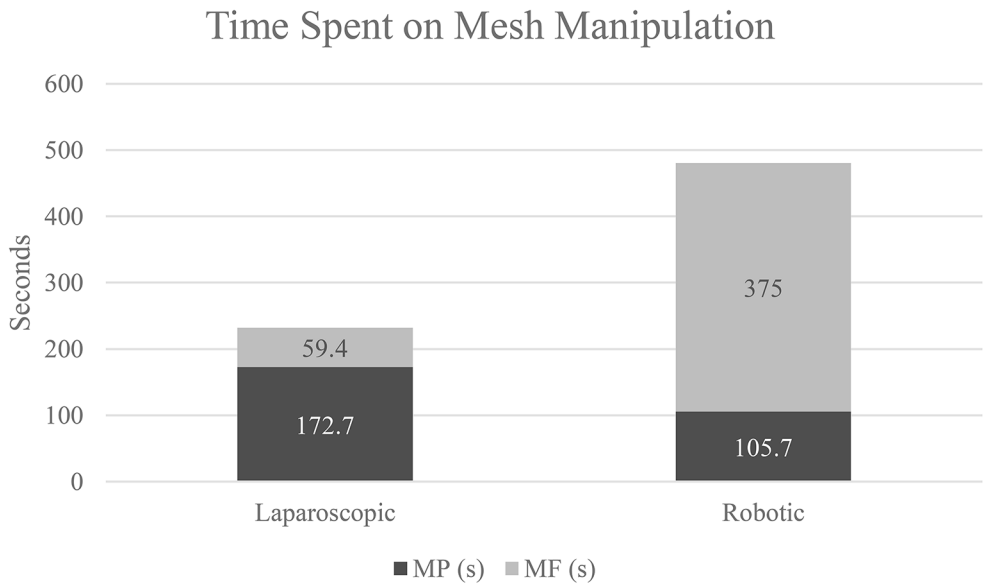

Fig. 1 Time (in seconds) spent on mesh placement and mesh fixation between laparoscopic and robotic cases. Abbreviations: MP = Mesh Placement; MF = Mesh Fixation

activation for the right AD, left ED, or left AD (Fig. 2a). During MF, the average $\%MVC_{RMS}$ was significantly higher for the LIHR group compared to the RIHR group in the right ED (42.7 vs. 18.8, $p < 0.0001$), right UT (24.5 vs. 9.0, $p = 0.0004$), right FCR (9.2 vs. 4.1, $p < 0.0001$), and left UT (23.6 vs. 8.5, $p = 0.0011$). There were no significant differences in muscle activation in the other four muscles investigated (Fig. 2b).

There was no significant difference in overall muscle fatigue between MP and MF ($p = 0.6847$). The combined analysis of MP + MF data showed a difference in muscle fatigue such that the LIHR group had lower fatigue compared to the RIHR group in the right ED (116.2 vs. 90.6, $p < 0.0001$), right AD (71.4 vs. 59.3, $p < 0.0001$), and left FCR (91.0 vs. 78.7, $p = 0.0001$). There were no significant differences in the remaining muscle groups (Fig. 3).

RULA and NASA-TLX data

The RULA tool was used to assess surgeon ergonomics in the operating room. A significantly higher score was seen in the right upper limb composite score (score C) in the LIHR group compared to the RIHR group (5.7 vs. 5.1, $p = 0.02$) for MP. A similar pattern was seen for the left upper limb composite score (score C) during MP and this difference was marginally significant (5.9 vs. 5.2, $p = 0.054$). Conversely, for MF, the RIHR group had a higher composite upper limb score (score C) for both the right (5.4 vs. 5.9) and the left (5.4 vs. 5.7) sides, but these results were not significantly different ($p = 0.07$ and 0.17, respectively). There were no differences between RULA grand total scores for either MP or MF (Table 2). We found no significant differences in NASA-TLX data between LIRH and RIHR ($p > 0.05$).

Association between EMG data and RULA scores

A generalized linear mixed model approach was applied to determine which EMG data, including %MVC and Fmed measurement, better estimated RULA scores. Muscle fatigue measurements (Fmed) offered the best model fit statistics compared to muscle activation ($\%MVC_{RMS}$), accounting for 38% of the variance in grand total RULA score and 28% of the variance in composite upper limb score.

Laterality (left vs. right side) for muscle fatigue measurements (Fmed) emerged as a predictor of RULA grand total score when interacting with surgical approach and three of the four muscle groups studied: UT, FCR, and ED. Specifically, lower muscle fatigue in the left UT in the RIHR group significantly reduced the RULA grand total left score (Simple slope = -3.90, $p = 0.009$). Lower muscle fatigue in the left FCR in the RIHR group also reduced the RULA grand total left score (simple slope = -3.49, $p = 0.008$). Conversely, in the right FCR, there was a trend that lower fatigue for this muscle was associated with increased RULA grand total right score in the LIHR group, but this was not significant after adjustment for multiple comparisons ($p = 0.144$; unadjusted $p = 0.036$). Overall, muscle fatigue in the ED was also a significant predictor of RULA grand total score, regardless of muscle laterality or operative approach, such that lower fatigue in ED muscles increased RULA total score (simple slope = 0.88, $p = 0.035$). Specifically in the left ED, lower fatigue was associated with higher RULA grand total left score in the RIHR group (simple slope = 2.42, $p = 0.009$) and lower fatigue in the right ED trended towards an increased RULA grand total right score in the LIHR group, but this was not statistically significant after multiple comparison adjustments (simple slope = 1.61, $p = 0.13$; unadjusted $p = 0.033$). Neither the left nor right

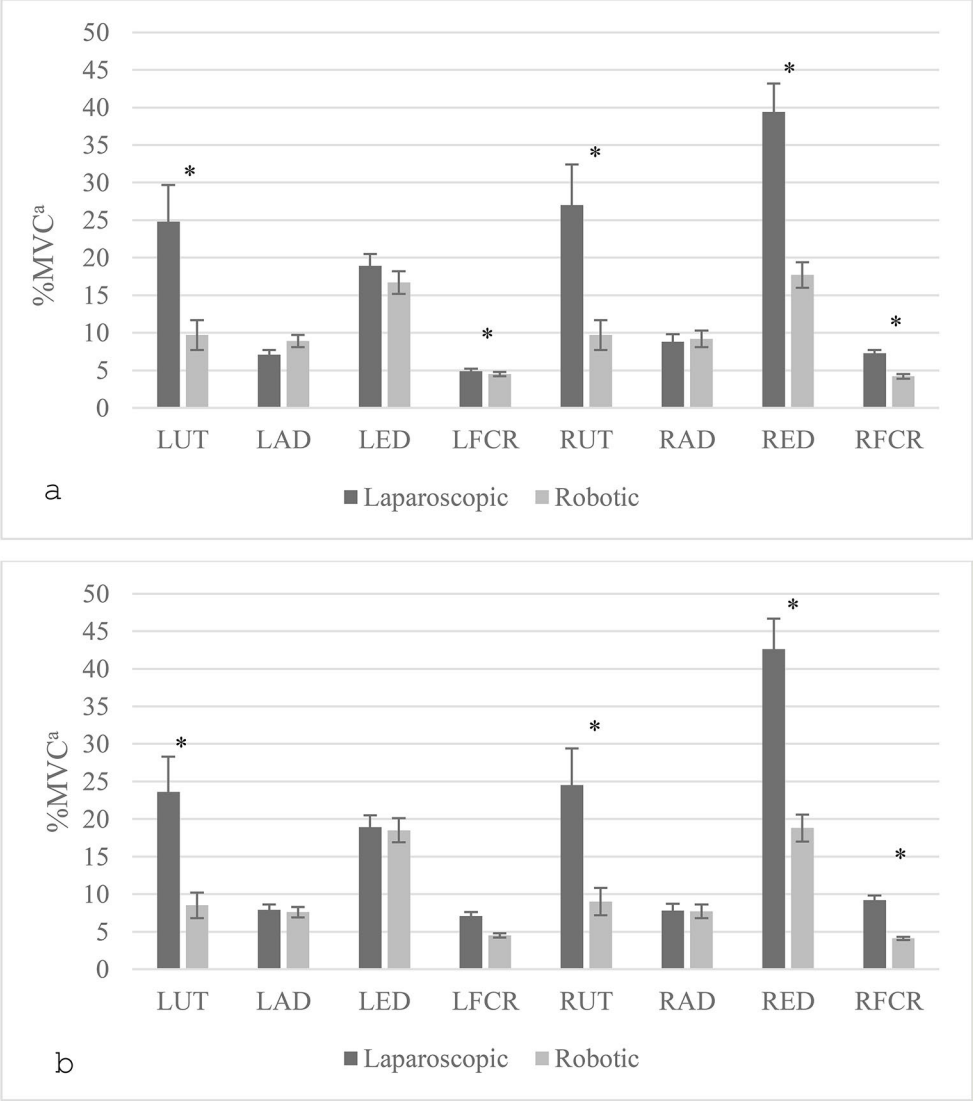

Fig. 2 **a-b** Muscle activation as a percent of maximum voluntary contraction. **a**) Muscle activation in each muscle for mesh placement. **b**) Muscle activation in each muscle for mesh fixation. Abbreviations: %MVC = Percent of maximum voluntary contraction; UT = upper trapezius, AD = anterior deltoid, ED = extensor digitorum, FCR = flexor carpi radialis; L = left, R = right. [a]Higher %MVC indicates higher muscle activation during the task as a percentage of maximum voluntary contraction in each individual muscle $^{*}p < 0.05$

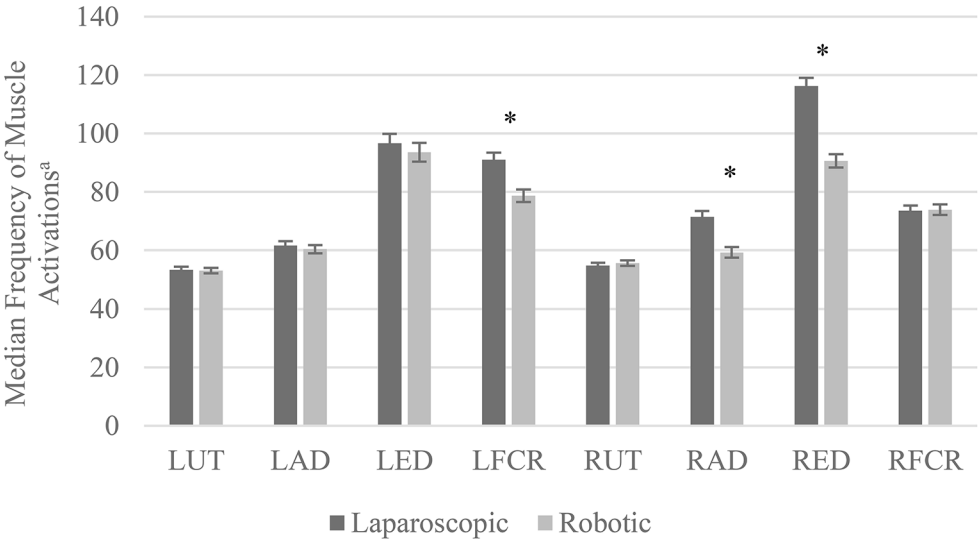

Fig. 3 Muscle fatigue as measured by median frequency of muscle activations. Abbreviations: UT = upper trapezius, AD = anterior deltoid, ED = extensor digitorum, FCR = flexor carpi radialis; L = left, R = right. [a]Higher median frequency of muscle activations (Fmed) indicates lower fatigue in each muscle. $^{*}p < 0.05$

Table 2 Operating surgeon rapid upper limb assessment score[a]

Rapid Upper Limb Assessment	Mean (SD)		
	Laparoscopic ($n=15$)	Robotic ($n=15$)	p Value
Mesh Placement			
Upper Limb Score Right (Score C)	5.7 (0.70)	5.1 (0.70)	0.02*
Upper Limb Score Left (Score C)	5.9 (0.88)	5.2 (0.94)	0.054
Lower Body Score (Score D)	2.9 (1.16)	3.3 (0.70)	0.16
Grand Score Right	4.8 (0.94)	4.5 (0.99)	0.51
Grand Score Left	4.7 (0.96)	4.6 (1.18)	0.76
Mesh Fixation			
Upper Limb Score Right (Score C)	5.4 (0.63)	5.9 (0.74)	0.07
Upper Limb Score Left (Score C)	5.4 (0.91)	5.7 (0.46)	0.17
Lower Body Score (Score D)	3.9 (2.1)	3.1 (0.46)	0.82
Grand Score Right	4.9 (1.36)	5.0 (0.85)	0.38
Grand Score Left	4.9 (1.30)	4.8 (0.56)	0.61

[a]A single evaluator subjectively scored muscle posture, use, and force for each side of the body separately and a grand total score is calculated. RULA grand scores range from 1–7 with higher scores indicating higher risk for musculoskeletal injury. Scores of 1–2 = negligible risk, 3–4 = low risk, 5–6 = medium risk, and 7 = imminent risk and should prompt immediate investigation and intervention

*$p<0.05$

AD emerged as a significant predictor of RULA grand total score for either operative approach (Fig. 4a-d).

Discussion

While minimally invasive surgical techniques have many advantages with respect to postoperative pain and recovery time, they have also been shown to increase surgeon fatigue and injury rates [4]. Understanding the patterns of injury during operations is an important first step towards making meaningful changes to improve surgeon ergonomics and reduce the burden of surgeon injury. Inguinal hernia repair is one of the most common operations performed globally and proper mesh placement and fixation are crucial for long-term durability of the repair [1–3, 12]. Because of the commonality of this case, IHR also facilitates ease of studying surgeon ergonomics. To our knowledge, this study is the first to investigate differences in surgeon ergonomics for specific steps in inguinal hernia repair. This study investigated ergonomics in mesh placement and fixation for LIHR and RIHR, measured via EMG monitoring of four upper extremity muscles bilaterally and the RULA tool. Overall, the results of the EMG analysis identified increased muscle activations in LIHR compared to RIHR for both mesh placement and fixation. Conversely, there was a general decreased muscle fatigue during LIHR compared to RIHR. Grand total RULA scores for both tasks were no different between the two approaches. Finally, muscle fatigue emerged as a better predictor of RULA grand total scores compared to muscle activation.

Previous studies have evaluated intraoperative ergonomics using EMG data, with similar findings that different muscle groups are preferentially utilized in laparoscopic and robotic-assisted operations [11, 16, 24]. Our study found overall higher levels of muscle activation in laparoscopic cases compared to robotic-assisted cases, with significantly higher activation in the left and right UT, right ED, and right FCR for both MP and MF and in the left FCR during MP. One could postulate that this is due to the haptic feedback present during LIHR, such that the surgeon force necessary for MP and MF is directly related to the patient's tissue integrity, compared to RIHR, which essentially eliminates this haptic feedback. A recent study also found higher muscle activation in laparoscopic operating platforms when completing simulated Fundamentals of Laparoscopic Skills tasks for both novice and expert operators, although the differences between laparoscopic and robotic-assisted cases were less pronounced during the experts' performances [11]. Although muscle activation was overall higher in LIRH, muscle fatigue was found to be lower, and the time spent manipulating mesh was shorter compared to RIHR. It is possible that these shorter periods of higher muscle activation are ergonomically protective compared to longer periods, although further investigation would be warranted to confirm this hypothesis.

In fact, only FCR had a lower average muscle activation than the deltoid muscles in our study. In contrast, previous work by Armijo et al. found higher muscle activation in bilateral UT and FCR for robotic-assisted cases compared to laparoscopic cases [16]. The authors also investigated measures of muscle fatigue and found that ED muscles had higher fatigue in laparoscopic cases, but that the remaining three muscle groups had similar fatigue regardless of surgical approach. In our study, we found decreased fatigue in the left FCR, right ED, and right AD in laparoscopic cases.

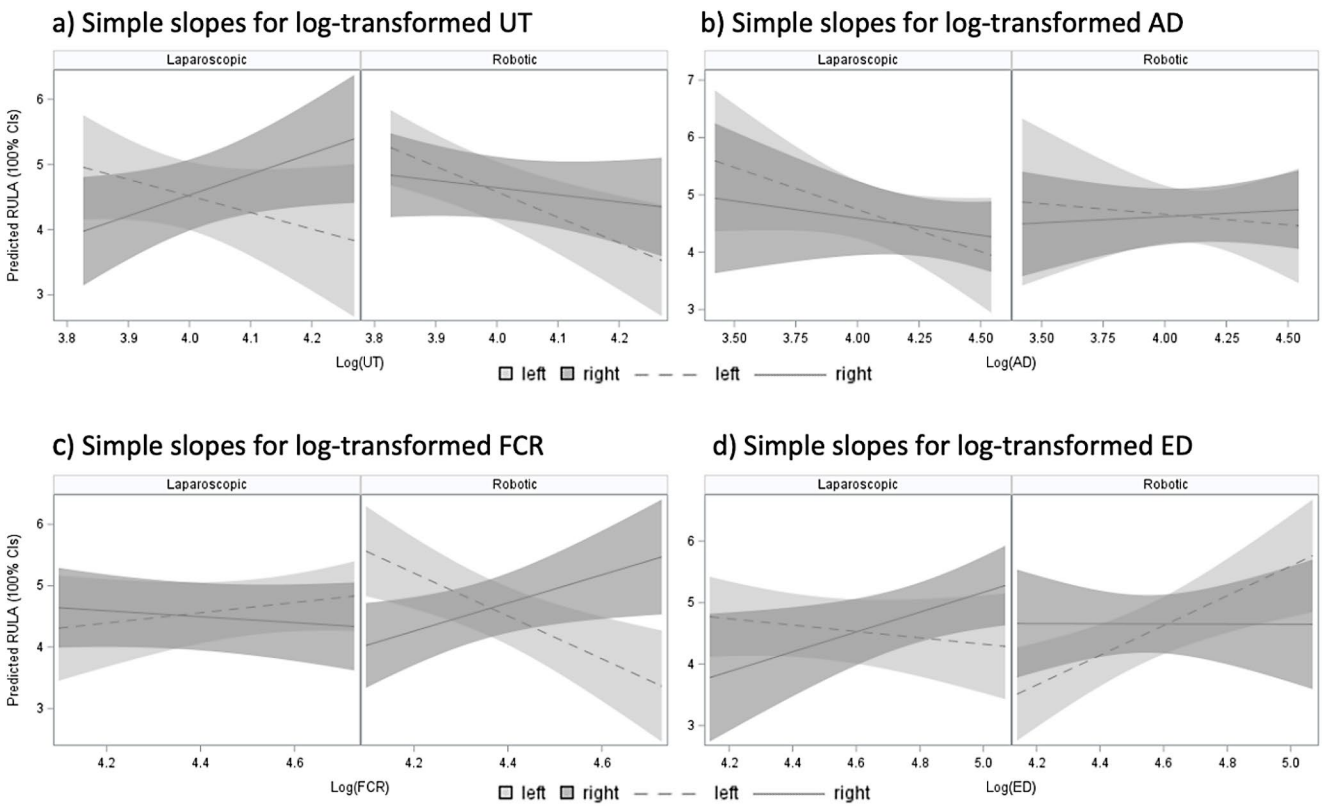

Fig. 4 **a-d** Generalized Linear Mixed Model for Muscle Fatigue and RULA scores[a]. An upward slope indicates a positive association while a downward slope indicates a negative association between muscle fatigue and RULA scores. **a**) Association between Muscle Fatigue and Predicted RULA scores for UT. **b**) Association between Muscle Fatigue and RULA scores for AD. **c**) Association between Muscle Fatigue and Predicted RULA scores for FCR. **d**) Association between Muscle Fatigue and Predicted RULA scores for ED. Abbreviations: UT = upper trapezius, AD = anterior deltoid, ED = extensor digitorum, FCR = flexor carpi radialis; L = left, R = right; RULA = Rapid Upper Limb Assessment.

[a] The effect plots display the predicted RULA total scores by the regression model controlling other variables with its mean or reference level (log_UT = 3.99. log_AD = 4.14, log_FCR = 4.37, log_ED = 4.59*)*. Using plot A as an example, in LIHR, increased muscle fatigue in the right UT was associated with higher RULA scores, while increased fatigue in the left UT was associated with lower RULA scores. In RIHR, decreased muscle fatigue was associated with decreased RULA scores for both the right and left UT muscles. Statistically significant interactions are discussed in the results section. The shaded areas represent 95% confidence intervals for each regression model

However, Armijo et al. evaluated a variety of operations from six surgical disciplines rather than limiting to one single procedure. Additionally, they evaluated each operation in its entirety rather than focusing on critical steps or portions of a case. Therefore, it is possible that the differences in muscle activation and fatigue findings in our study are explained by different muscle activation patterns between operative steps.

There were no differences in grand total RULA scores between approaches in this study. The RULA tool has been used as an ergonomic measure in other studies focusing on surgeon posture. The RIVAL trial investigated multiple measures of patient and surgeon factors during laparoscopic and robotic-assisted inguinal hernia repairs and also found no difference in total RULA scores, indicating, at least for this particular operation, that the risk for musculoskeletal injury in similar between approaches [14]. Regardless, the risk of injury as measured by our RULA for both laparoscopic and robotic-assisted cases fall closest to the medium risk category (Grand Score of 5–6), warranting investigation and possible intervention should be considered to protect surgeons from imminent risk of injury. Some surgical societies and independent investigators have begun to increase awareness of surgeon ergonomics and implement training programs to reduce fatigue and strain both in and out of the operating room [25–28]. However, to our knowledge, none of these interventions are designed to address muscle demands for a specific operation or operative approach. We are hopeful that the findings from our study serve as a foundation for future studies to investigate specific steps of an operation that may be more ergonomically riskier than others, which could lead to targeted interventions to further reduce the risk of surgeon injury.

Although both LIHR and RIHR show equivalent risk of injury based on the RULA tool, there were different muscle demands between the two operations. For instance, in this study, the time spent on MP + MF was shorter for LIHR compared to RIHR. An experienced surgeon may be able to efficiently complete a task without causing undue muscle fatigue. It is possible that our LIHR surgeon has adopted an operative strategy that maximized muscle activation in order to reduce time to task completion and, therefore, decrease muscle fatigue. This may be supported by previous studies that have found differences in expertise for surgical tasks can affect muscle activation patterns such that those who are more experienced have lower muscle activation based on surgeon experience [11].

While we did not examine differences in every individual component of the upper limb scores for the RULA, the upper arm posture category does include a component for increasing the score based on shoulder position, which may have contributed to an overall increase in score for this body region based on surgeon positioning to achieve appropriate MP and MF. Perhaps the ergonomics of standing at the bedside during MP and sitting at the console during MF lends itself to increased elevation of the shoulders during minimally invasive inguinal hernia repair to successfully complete these tasks. This hypothesis is supported by the study from Dalager et al. who studied surgeon posture between laparoscopic and robotic operations and found that robotic surgery was overall less demanding on a surgeon's posture than laparoscopic surgery, especially with regard to long-term static muscle activity and overall physical workload during the operation [29]. If so, there is the potential for development of targeted interventions to help surgeons become more aware of these deviations from safe ergonomics to help mitigate muscle injury. While this would be the goal in the long run, it should be noted that the process of changing one's posture to be more ergonomically correct during an operation may initially lead to increased discomfort and fatigue due to the need to re-train the muscles to work together differently. This highlights the importance of surgeons understanding the ergonomic risks of operations early in their careers to avoid the development of muscle overuses that cause undue strain on them.

This study is not without limitations. First, this was a single-center study with a small sample size utilizing four fellowship-trained surgeons, specifically studying two shoulder and two forearm muscle groups. Therefore, results may not be generalizable to other surgeons, operative environments, or other important muscles such as the postural muscles of the cervical, thoracic, or lumbar spine. Second, there was a lack of randomization between groups. Third, only one surgeon performed the cases laparoscopically (via TEP) while three surgeons performed the robotic-assisted operations (via TAPP). It is possible that the differences in muscle activation and fatigue are based on inter-group variations or procedural variations in TEP versus TAPP hernia repairs, rather than inherent differences between laparoscopic inguinal hernia repair and robotic-assisted inguinal hernia repair. However, we specifically wanted each surgeon to operate via the approach they use on a regular basis in order to capture more true-to-life data regarding their ergonomics. Therefore, randomization was intentionally avoided. While avoidance of randomization may have introduced other factors that affect ergonomics, such as the weight of the patient or the size of the inguinal hernia, operating times (specifically leading up to the studied task), which often serves as a surrogate for case complexity, were similar between the LIHR and RIHR groups. Therefore, we suspect that these additional factors were similarly present in both groups [30]. Additionally, measures were taken during analysis to control for differences between surgeons whenever possible. Finally, our intent was to evaluate the ergonomics of MP and MF only. Therefore, differences in ergonomics for the other critical steps of minimally invasive inguinal hernia repair cannot be inferred from this study.

In conclusion, this study found significant differences in muscle utilization patterns in MP and MF for minimally invasive inguinal hernia repairs, including increased activation and decreased fatigue in the upper extremity muscles in laparoscopic cases compared to robotic-assisted cases. Despite differences in EMG measurements, there was no significant difference in total RULA scores between approaches and RULA scores indicated a moderate risk of musculoskeletal injury associated for MP and MF in both groups. While this data is very granular, we believe that identifying physiologic differences in muscle usage between surgical approaches is a key initial step to identifying and mitigating risk of injury in order to develop interventions targeted at preserving a surgeon's career longevity. Future studies are needed to validate our findings on a larger scale. Additionally, the identification of ergonomically riskier steps of more surgical operations may enhance future efforts to protect surgeons against physical harm or muscle overuse.

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Declarations

Ethical approval and informed consent This study was approved by the Institutional Review Board (IRB) at the University of Nebraska Medical Center (Protocol # 572-19-EP). Informed consent was obtained from each surgeon prior to enrollment in the study.

Conflict of interest The authors declare that they have no conflicts of interest to disclose regarding the content of this study.

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